

(a) Candidate windows



onset

20-90 s



energy end



1%-99% energy



(b) Amplitude-energy audit

Peak ratio ≥ 0.99

Energy ratio ≥ 0.95

Peak times inside

Keep the shortest passing window

(c) Spectral safeguard

PSA (0.2 s) ≥ 0.95

PSA (1.0 s) ≥ 0.95

PSA (3.0 s) ≥ 0.95

Fail: 1%-99% energy window

Fail again: full record

(d) Product window

 Primary window

 Energy-window escalation

 Full-record fallback

Record-level audit trail